

SINUJU, Korea, Democratic People's Republic of

WMO: 470350

Lat: 40.100N

Lon: 124.383E

Elev: 23

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.0	7.6	-12.3	2.8	10.8	-8.3	3.5	12.5	18.9	16.3	16.4	17.4	5.4	0	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	88.4	75.8	85.7	74.6	83.5	73.5	79.2	84.5	77.7	82.3	76.3	80.5	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.7	144.3	82.3	76.4	137.8	80.5	75.0	131.4	79.0	42.9	85.1	41.3	82.3	39.9	80.7	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.2	13.5	11.5	-1.0	92.8	4.8	3.2	-4.4	95.2	-7.2	97.0	-9.9	98.9	-13.4	101.2	
			DB												
			WB	-1.8	81.9	4.6	2.6	-5.1	83.8	-7.8	85.3	-10.4	86.8	-13.7	88.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	20.7	27.1	38.1	50.6	61.5	70.1	76.1	76.7	68.1	55.1	38.8	24.9	(6)
(7)		DBStd	20.43	7.21	7.08	6.14	5.90	4.71	4.08	3.56	3.74	4.82	6.65	8.76	7.33	(7)
(8)		HDD50	3142	908	642	373	63	0	0	0	0	30	346	780		(8)
(9)		HDD65	6203	1373	1062	835	432	127	7	0	25	311	786	1245		(9)
(10)		CDD50	3420	0	0	2	80	358	603	809	828	543	187	10	0	(10)
(11)		CDD65	1007	0	0	0	0	20	159	344	363	118	3	0	0	(11)
(12)		CDH74	6303	0	0	0	6	146	790	2145	2678	527	11	0	0	(12)
(13)		CDH80	1514	0	0	0	0	18	164	555	721	55	1	0	0	(13)
(14)	Wind	WSAvg	4.3	4.7	4.6	4.7	4.6	4.3	3.7	3.5	3.7	3.9	4.3	4.7	4.8	(14)
(15)	Precipitation	PrecAvg	38.1	0.4	0.4	0.7	2.1	2.6	4.4	11.3	9.1	3.7	1.7	1.1	0.5	(15)
(16)		PrecMax	58.0	2.4	1.6	2.1	5.7	5.7	10.5	25.7	20.0	7.7	6.7	2.4	1.3	(16)
(17)		PrecMin	26.4	0.0	0.0	0.1	0.3	0.4	0.8	3.9	1.5	0.7	0.0	0.2	0.0	(17)
(18)		PrecStd	8.3	0.4	0.4	0.5	1.4	1.5	2.5	5.7	5.0	1.9	1.5	0.7	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	49.0	60.4	73.4	82.4	88.0	92.5	92.4	84.4	75.4	61.8	45.5	(19)
(20)		0.4%	MCWB	35.0	39.3	47.1	57.5	64.4	70.9	79.1	78.8	71.8	63.1	54.5	41.8	(20)
(21)		2%	DB	37.1	43.5	54.3	68.1	77.8	84.2	88.6	88.7	81.4	70.9	57.8	41.2	(21)
(22)		2%	MCWB	32.8	36.2	43.9	53.7	62.0	68.9	76.3	76.8	69.1	61.1	51.6	37.3	(22)
(23)		5%	DB	34.2	40.1	50.8	64.3	74.5	81.4	85.4	86.3	79.3	67.9	54.6	38.1	(23)
(24)		5%	MCWB	29.9	34.8	41.9	51.9	60.7	68.4	75.3	75.5	68.0	58.9	49.2	34.3	(24)
(25)		10%	DB	31.4	37.6	47.6	61.1	71.5	78.5	82.8	84.0	77.2	65.2	51.6	35.4	(25)
(26)		10%	MCWB	27.4	33.0	40.3	50.2	59.7	68.2	74.5	74.9	67.2	57.6	46.9	31.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	40.7	49.9	60.4	68.2	74.1	81.7	81.9	75.3	66.2	57.0	43.0	(27)
(28)		0.4%	MCDWB	39.0	45.7	56.6	68.1	77.0	82.5	89.3	87.8	80.5	71.9	60.0	44.8	(28)
(29)		2%	WB	33.3	37.7	46.0	56.6	65.1	72.1	79.2	79.7	73.1	63.5	53.3	37.8	(29)
(30)		2%	MCDWB	36.5	41.8	51.2	63.8	73.3	79.5	85.2	84.6	78.0	68.5	56.9	40.3	(30)
(31)		5%	WB	30.5	35.7	43.7	54.3	63.2	70.9	77.4	78.2	71.3	61.5	50.3	35.0	(31)
(32)		5%	MCDWB	33.9	39.6	48.5	61.3	70.8	77.3	82.2	82.9	76.4	65.9	53.6	37.6	(32)
(33)		10%	WB	27.4	33.1	41.4	52.1	61.5	69.7	76.1	77.1	69.6	59.4	47.3	31.9	(33)
(34)		10%	MCDWB	31.0	37.1	46.6	59.0	68.6	75.6	80.1	81.5	74.7	63.5	50.8	34.8	(34)
(35)	Mean Daily Temperature Range		MDBR	14.6	15.3	15.2	16.3	16.4	13.5	10.4	12.7	16.3	16.6	13.8	13.0	(35)
(36)		5% DB	MCDBR	16.2	17.5	19.3	21.4	22.0	19.5	16.1	16.4	18.2	18.9	16.8	14.7	(36)
(37)			MCWBR	12.9	12.7	11.8	10.7	10.3	8.5	6.8	7.4	8.9	10.9	12.3	12.2	(37)
(38)			MCDBR	15.4	15.9	17.2	18.0	18.1	14.9	12.4	12.3	14.4	15.9	14.9	13.9	(38)
(39)		5% WB	MCWBR	13.0	12.7	12.1	10.7	10.0	7.6	6.5	7.2	8.6	11.0	12.3	12.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.401	0.479	0.540	0.627	0.654	0.736	0.711	0.605	0.531	0.558	0.489	0.415	(40)	
(41)		taud	1.959	1.808	1.730	1.609	1.604	1.522	1.612	1.817	1.915	1.759	1.851	1.949	(41)	
(42)		Ebn at Noon	224	221	224	215	213	196	200	218	225	195	190	206	(42)	
(43)		Edh at Noon	41	56	67	80	82	90	81	65	55	58	46	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	765	1004	1368	1579	1768	1561	1320	1425	1401	1048	731	651	(44)	
(45)		RadStd	49	71	98	124	144	191	140	166	115	66	76	50	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-3.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page